

Initialization, Normalization, and Debugging

Scaler School of Technology | Deep Learning I | Week 3 — Session 2/2 | 07 May 2026



What Is This Lecture About?

A well-designed architecture is useless if training fails. This session teaches you **why** networks sometimes refuse to learn (vanishing/exploding gradients), **how** to set up weights so training starts smoothly (initialization), **how** to stabilize activations throughout training (normalization), and **how** to systematically debug when things go wrong.

1 Vanishing/Exploding Gradients & Initialization

Why multiplying many small (or large) numbers kills gradient flow in deep networks, and how Xavier and He initialization keep signals stable. *Like the telephone game — messages degrade with every relay unless you amplify correctly.*

2 Batch Normalization & Layer Normalization

Normalizing activations inside the network to stabilize training. BatchNorm normalizes across the batch (great for vision); LayerNorm normalizes across features (essential for Transformers). *Like standardizing ingredients before cooking — consistent inputs produce consistent results.*

3 Debugging Training Runs

Gradient checking, recognizing common failure modes (loss = NaN, loss stuck at $\ln(k)$), and a systematic debugging checklist that saves hours of trial-and-error.

What to Expect in the Session

- Live demo: watching activations vanish vs. stay stable
- Implementing BatchNorm from scratch
- ★ The **debugging checklist** that saves hours of frustration
- Comparing BatchNorm vs LayerNorm axes visually
- 2 quizzes + a “find the 3 bugs” challenge
- Gradient checking: verifying your backprop is correct

How to Prepare (Optional but Helpful)

- ✓ **Recall** the chain rule from backpropagation (Session 3) — gradients are products of partial derivatives
- ✓ **Recall** the BatchNorm recap from last session — we'll go deeper into its internals
- ✓ **Think about:** “If you multiply 0.9 by itself 50 times, what do you get? What about 1.1?”
- ✓ **Skim** PyTorch docs for `nn.BatchNorm1d` and `nn.LayerNorm` if curious

Duration: ~2 hours
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Prerequisites: Backprop, chain rule, basic regularization (Session 5)

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